

BUSINESS ANALYTICS PROJECT

Business Analytics for Supermart Grocery Sales and Profit Analysis

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Project documentation completed using the supplied project notebook and the supplied Applied Business Project format.

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1. Introduction

What is Business Analytics?

Business Analytics is the use of data, statistics and analytical tools to understand business performance and support better decision-making.

Why is data important?

Data helps a business understand sales performance, customer purchasing patterns, profitable products, regional performance and future trends.

Business area selected

Retail – Supermarket Grocery Sales

Why this business?

A supermarket generates a large number of sales transactions involving different product categories, regions, discounts and profit values. Therefore, it is suitable for applying Business Analytics techniques.

How can analytics help?

Analytics can help the supermarket identify:

- Best-performing product categories
- High-sales regions
- Profitable products
- Effect of discounts on profit
- Sales trends over time
- Future sales patterns
- Business opportunities and risks

2. Problem Statement

The objective of this project is to analyze Supermart grocery sales data to understand sales performance, profitability, discount patterns, regional performance and time-based sales trends.

Problem:

The supermarket needs to identify which product categories and regions generate the highest sales and profit and understand the relationships between Sales, Discount and Profit.

Data available:

9,994 supermarket sales transactions containing 11 variables.

Analysis required:

- Sales and profit analysis
- Category and regional analysis
- Discount analysis
- Correlation analysis
- Regression analysis
- Probability analysis
- Forecasting
- Statistical testing
- Business decision-making

Decisions:

- Which products should receive more attention?
 - Which regions should be prioritized? • How should discounts be managed?
 - How can future sales be planned?
 - How can profitability be improved?
-

3. Objectives

- To collect and understand supermarket sales data.
- To perform data preprocessing and exploratory data analysis.
- To analyze sales and profit across product categories.
- To study the relationship between Sales, Discount and Profit.
- To perform regression analysis for business prediction.
- To analyze sales trends over time.
- To perform statistical testing and probability analysis.
- To develop a Business Intelligence dashboard.
- To identify important business insights.
- To support data-driven business decisions.

4. Business Scenario / Domain Description

Business Name

Supermart Grocery Store

Type of Business

Retail grocery supermarket.

Products

The dataset contains the following major categories:

- Beverages
- Food Grains
- Fruits & Veggies
- Bakery
- Snacks
- Eggs, Meat & Fish
- Oil & Masala

Target Customers

The supermarket serves general grocery customers purchasing food, beverages and household grocery products.

Business Operations

The business records:

- Customer purchases
- Product categories
- Sub-categories
- Sales
- Discounts

- Profit
- Cities
- Regions
- Order dates

Important Business Problems

The supermarket needs to:

- Increase sales
- Improve profitability
- Identify high-performing categories
- Manage discounts
- Identify strong and weak regions
- Forecast future sales

5. Dataset Description

Dataset Name: Supermart Grocery Sales Dataset

Dataset Source: The CSV dataset used for this project. Add the original Kaggle or download URL here after verifying the source.

Number of Records: 9,994

Number of Columns: 11

The dataset contains information about orders, customers, product categories, sub-categories, cities, order dates, regions, sales, discounts and profit. It is suitable for analyzing retail sales performance and profitability.

Column Description

Column	Description	Data Type
Order ID	Unique order identifier	Categorical
Customer Name	Customer name	Categorical
Category	Main product category	Categorical
Sub Category	Product sub-category	Categorical
City	Customer/order city	Categorical
Order Date	Date of order	Date/Time
Region	Sales region	Categorical
Sales	Sales amount	Numerical

Discount	Discount percentage	Numerical
Profit	Profit generated	Numerical
State	State of operation	Categorical

Numerical Variables

- Sales
- Discount
- Profit

Categorical Variables

- Order ID
- Customer Name
- Category
- Sub Category
- City
- Region
- State

Date/Time Variable

- Order Date

6. Data Collection

Data Source

The project uses the supplied **Supermart Grocery Sales – Retail Analytics Dataset** CSV file.

Collection Method

The dataset contains supermarket transaction records representing grocery sales.

Number of Records

9,994

Time Period

The dataset contains orders from approximately **2015 to 2018**.

Variables Collected

The dataset contains order details, customer names, product categories, sub-categories, cities, regions, sales, discounts and profit.

7. Data Preprocessing

The following preprocessing operations were performed:

1. Missing Values

The dataset was checked using:

```
df.isnull().sum()
```

Result:

0 missing values were found in all columns.

2. Duplicate Records

The dataset was checked using:

```
df.duplicated().sum()
```

Result:

0 duplicate records were found.

3. Data Types

The dataset initially contained Order Date as an object/string field.

It was converted into DateTime format using:

```
df['Order Date'] = pd.to_datetime(df['Order Date'], errors='coerce')
```

4. Invalid Values

Sales, Discount and Profit values were checked for valid numerical values.

No missing numerical values were found.

5. Date Formatting

The dataset contains mixed date formats. The Order Date column was converted into DateTime format to support monthly sales analysis and time-series analysis. The date conversion and resulting date values were checked before performing time-based calculations.

6. Final Dataset

The final dataset contains:

9,994 records and 11 columns.

8. Exploratory Data Analysis

The following EDA operations were performed: **First Five Records**

```
df.head()
```

Last Five Records

```
df.tail()
```

Dataset Shape

```
df.shape
```

Output:

(9994, 11)

Column Names

```
df.columns
```

Data Types

```
df.dtypes
```

Missing Values

```
df.isnull().sum()
```

Result:

0 missing values

Duplicate Records

```
df.duplicated().sum()
```

Result:

0 duplicate records Unique

Values Important unique

values include:

- 7 product categories
- 23 sub-categories
- 24 cities
- 5 regions
- 50 customer names
- 1 state

Recommended Charts

The following charts should be included:

1. Sales by Category – Bar Chart
2. Profit by Category – Bar Chart
3. Sales by Region – Bar Chart
4. Sales Distribution – Histogram
5. Profit Distribution – Histogram
6. Sales – Box Plot
7. Discount Distribution – Histogram
8. Sales vs Profit – Scatter Plot
9. Category Sales – Pie Chart

Observation

Eggs, Meat & Fish has the highest total sales among the categories. Snacks has the highest total profit. West is the strongest region by sales. Sales and Profit show a positive relationship, while Discount has almost no linear relationship with either Sales or Profit.

9. Descriptive Statistical Analysis

The following statistics were calculated for Sales, Discount and Profit.

	Order Date	Sales	Discount	Profit
count	4042	9994.000000	9994.000000	9994.000000
mean	2017-04-28 03:16:17.931716864	1496.596158	0.226817	374.937082
min	2015-01-03 00:00:00	500.000000	0.100000	25.250000
25%	2016-05-09 06:00:00	1000.000000	0.160000	180.022500
50%	2017-07-01 00:00:00	1498.000000	0.230000	320.780000
75%	2018-06-01 00:00:00	1994.750000	0.290000	525.627500
max	2018-12-11 00:00:00	2500.000000	0.350000	1120.950000
std	NaN	577.559036	0.074636	239.932881

Interpretation

- Average Sales per transaction is approximately **₹1,496.60**.
- Median Sales is **₹1,498**, which is very close to the mean.
- Average Discount is approximately **22.68%**.
- Average Profit is approximately **₹374.94**.
- Maximum Profit is **₹1,120.95**.
- Sales has considerable variation with a standard deviation of **₹577.56**.

10. Distribution Analysis

Variable Selected

Sales

Results

- Mean Sales = ₹1,496.60
- Median Sales = ₹1,498
- Standard Deviation = ₹577.56
- Minimum = ₹500
- Maximum = ₹2,500

The Sales distribution is approximately balanced because the mean and median are very close.

The Sales skewness is approximately **0.001**, indicating almost no skewness.

Interpretation

Most sales transactions are distributed around the centre of the range. The difference between the minimum and maximum values shows variation in transaction values.

11. Correlation Analysis

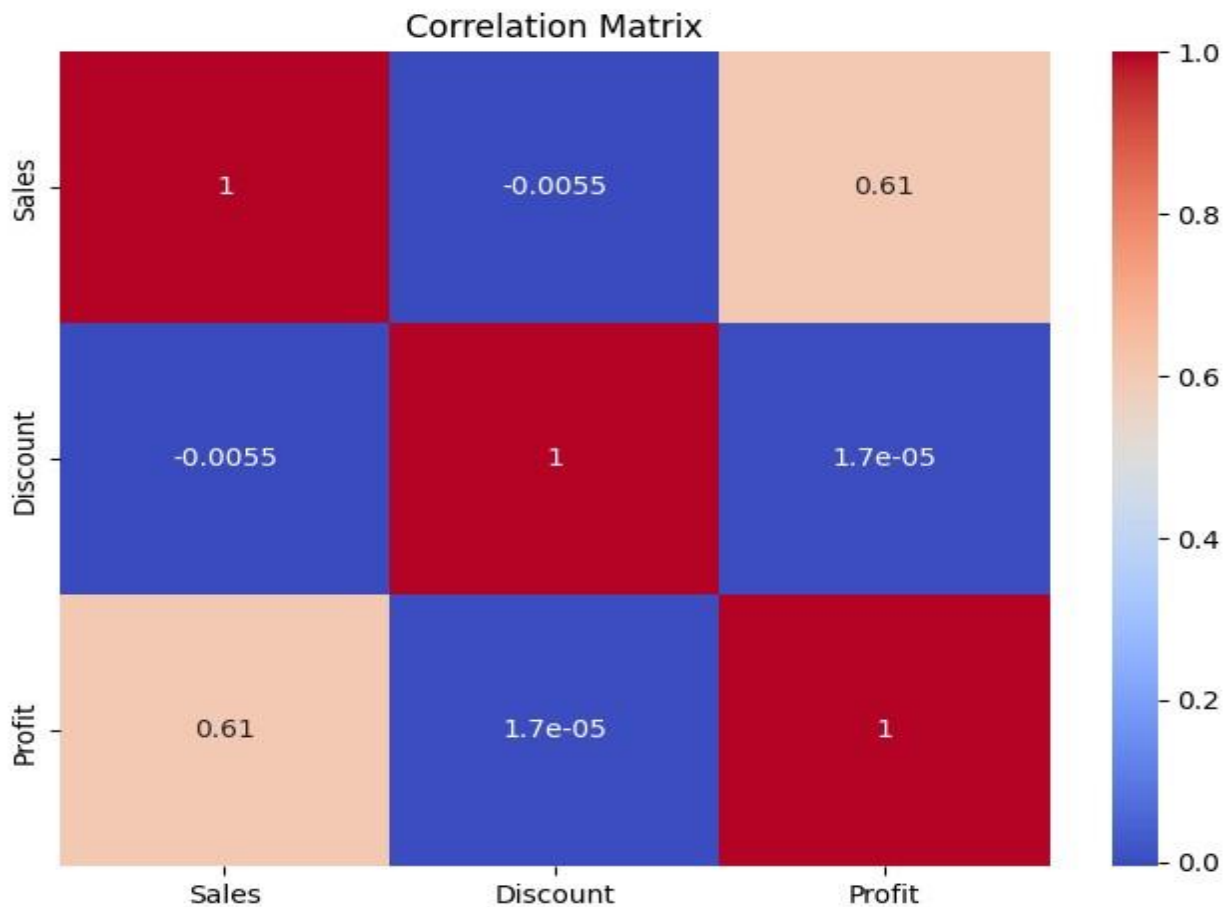
The correlation matrix for Sales, Discount and Profit is:

	Sales	Discount	Profit
Sales	1.000000	-0.005512	0.605349
Discount	-0.005512	1.000000	0.000017
Profit	0.605349	0.000017	1.000000

Interpretation

- Sales and Profit have a **moderately strong positive relationship (0.6053)**.
- Sales and Discount have an almost zero relationship (**-0.0055**).
- Discount and Profit have almost zero correlation (**0.0000**).
- Therefore, higher sales are generally associated with higher profit.
- Discount alone does not show a meaningful linear relationship with profit in this dataset.

Heatmap



12. Regression Analysis

Independent Variable

Sales

Dependent Variable

Profit

The following model was used:

```
X = df[['Sales']]
```

```
y = df['Profit']
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

Regression Equation

Profit = -1.423 + 0.2515 × Sales

Results

- Coefficient = **0.2515**
- Intercept = **-1.423**
- R^2 = **0.3664**

Prediction Example

The regression equation obtained from the model is:

$$\text{Profit} = -1.423 + 0.2515 \times \text{Sales}$$

For a Sales value of ₹1,500, the estimated Profit is approximately ₹376.83, based on the displayed regression equation.

Interpretation

The regression model indicates a positive relationship between Sales and Profit. The R^2 value of 0.3664 indicates that approximately 36.64% of the variation in Profit is explained by Sales in this model. Other variables may also influence Profit.

13. Business Intelligence Dashboard – Tableau

Dashboard Title

Supermart Grocery Sales and Profitability Dashboard

Objective

The dashboard is designed to monitor sales, profit, discount patterns, product category performance and regional sales performance.

Key Performance Indicators (KPIs)

1. Total Sales
2. Total Profit
3. Total Orders
4. Average Discount
5. Overall Profit Margin

Visualizations

1. Sales by Category – Bar Chart
2. Profit by Category – Bar Chart
3. Sales by Region – Bar Chart
4. Monthly Sales Trend – Line Chart
5. Sales vs Profit – Scatter Plot
6. Top 10 Products or Sub-Categories by Sales

Interactive Filters

- Region
- Category
- Sub-Category
- City
- Order Date

Calculated Field: Profit Margin

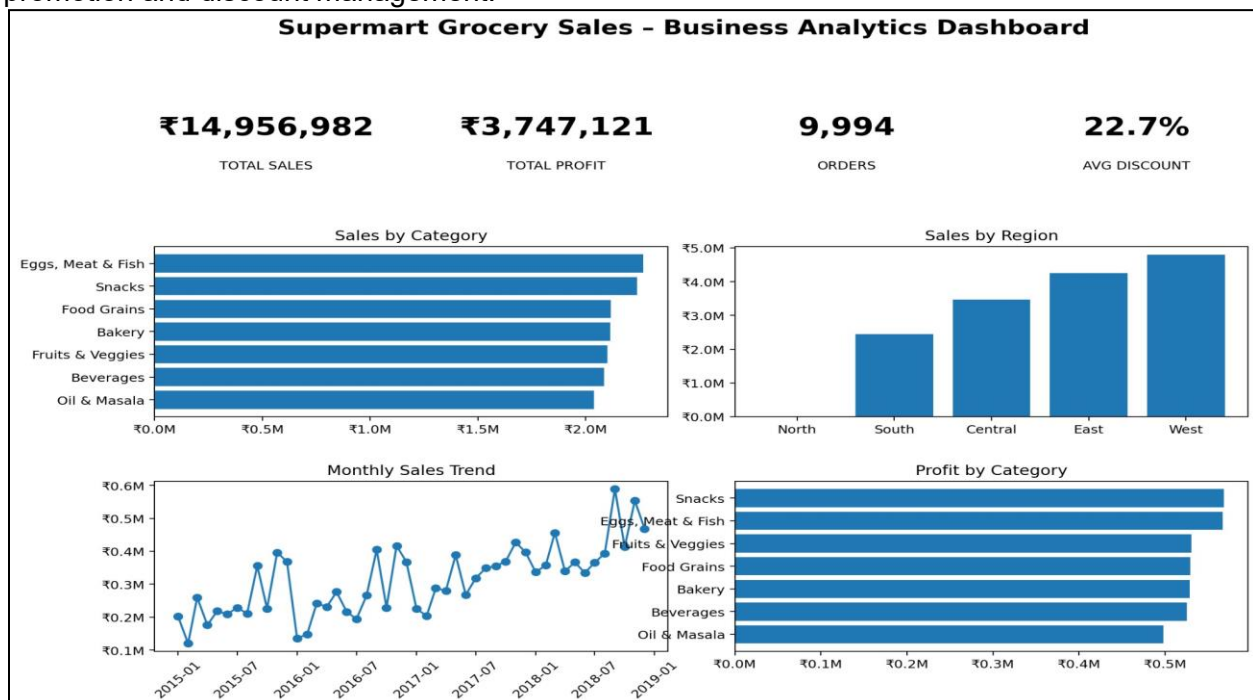
Create a calculated field using:

$SUM([Profit]) / SUM([Sales])$

Format the result as a percentage.

Expected Business Use

The dashboard can help users compare category and regional performance, monitor profitability and explore monthly sales trends. It can support data-driven decisions about inventory, product promotion and discount management.



14. Probability and Probability Distributions

Probability Experiment

The probability of a transaction having Sales greater than average Sales was calculated.

Average Sales:

₹1,496.60

Probability:

$$P(\text{Sales} > \text{Average Sales}) = 0.5008$$

Therefore, approximately **50.08%** of transactions have Sales above the average.

Other Useful Probabilities

● Probability Analysis

- The probability of a transaction having Sales greater than the average Sales was calculated using the dataset.
- Average Sales: Approximately ₹1,496.60 ● Probability:
- $P(\text{Sales} > \text{Average Sales}) = 0.5008$ ● Interpretation:
- Approximately 50.08% of the transactions have Sales greater than the average Sales value. This probability helps describe the distribution of transaction values in the dataset.

Distribution Examples

For business simulation, the following distributions can be used:

- Normal Distribution – approximately bell-shaped
- Uniform Distribution – values distributed evenly
- Exponential Distribution – right-skewed distribution

15. Decision Making Under Uncertainty

Business Decision

The supermarket needs to decide whether to run a **low-discount, medium-discount or high-discount promotional strategy**.

Alternative	High Demand	Medium Demand	Low Demand
Low Discount	+₹80,000	+₹50,000	+₹10,000
Medium Discount	+₹90,000	+₹55,000	+₹5,000
High Discount	+₹100,000	+₹50,000	-₹10,000

Assumed probabilities:

- High demand = 0.30
- Medium demand = 0.50
- Low demand = 0.20

Expected values:

- Low Discount = ₹51,000
- Medium Discount = ₹56,500

- High Discount = ₹52,000

Decision

The Medium Discount strategy has the highest expected value of ₹55,500 under the assumed demand probabilities.

This is an illustrative decision-making scenario. The payoffs and demand probabilities are assumptions and are not directly observed from the historical dataset.

16. Sampling and Sampling Distribution

Population

All **9,994** supermarket transactions.

Sample

Random sample of **30** transactions.

Sampling Technique

Simple Random Sampling.

Repetitions

100 random samples were generated.

Sample Size	Mean of Sample Means	SD of Sample Means
10	1515.80	185.76
20	1501.48	145.59
30	1506.51	114.51
40	1498.31	84.20

Interpretation

As sample size increases, the variability of sample means decreases.

This demonstrates the **Central Limit Theorem**: repeated sample means tend to form an approximately normal distribution around the population mean.

17. Confidence Interval Estimation

For the Sales variable:

90% Confidence Interval

₹1,487.09 to ₹1,506.10

95% Confidence Interval

₹1,485.27 to ₹1,507.92

99% Confidence Interval

₹1,481.71 to ₹1,511.48

Interpretation

The 95% confidence interval estimates the population's average Sales to be approximately between **₹1,485.27 and ₹1,507.92**.

A higher confidence level produces a wider interval.

18. Bootstrapping

Variable Selected

Sales

Original Median

₹1,498

Bootstrap Resampling

1,000 bootstrap samples were generated using sampling with replacement.

Results

- Mean Bootstrap Median = **₹1,497.43**
- 95% Bootstrap Confidence Interval = **₹1,480.49 to ₹1,516.00**

Interpretation

The bootstrap results are close to the original median, indicating that the estimated median Sales is reasonably stable.

19. Hypothesis Testing

Business Question

Is the average Sales significantly different from ₹1,500?

Hypotheses

H_0 : Mean Sales = ₹1,500

H_1 : Mean Sales $\neq$ ₹1,500 **Significance**

Level

$\alpha = 0.05$

Test

One-sample t-test.

Results

- t-statistic = **-0.589**
- p-value = **0.5558**
- $\alpha = 0.05$

Decision

Fail to reject H_0 .

Business Interpretation

There is no statistically significant evidence that the average Sales differs from ₹1,500.

Conclusion

The p-value is 0.5558, which is greater than the significance level of 0.05. Therefore, the null hypothesis is not rejected.

There is insufficient statistical evidence to conclude that the population mean Sales differs from ₹1,500.

20. A/B Testing

Since the dataset does not contain actual website conversion data, an **A/B-style comparison** can be performed using discount groups.

Group A

Transactions with Discount $\leq 23\%$.

Group B

Transactions with Discount > 23%.

Group	Records	Average Sales
A – Lower Discount	5,269	₹1,500.04
B – Higher Discount	4,725	₹1,492.76

Test Result

- t-statistic = **0.629**
- p-value = **0.5291**
- α = **0.05**

Decision

Fail to reject H_0 .

Business Interpretation

The observed difference in average Sales between lower- and higher-discount groups is not statistically significant.

Therefore, the dataset does not provide strong evidence that higher discounts alone increase Sales.

Note: This is an observational A/B-style comparison, not a randomized experiment.

Conclusion

The comparison did not find a statistically significant difference in average Sales between the lower-discount and higher-discount groups at the 5% significance level.

This is an observational comparison of existing transactions, not a randomized experiment. Therefore, the results do not establish that discount levels cause a change in Sales.

21. Time Series Analysis

The Order Date field was converted into DateTime format and Sales were aggregated by month.

Time Period

Approximately **2015–2018**

Time-Series Pattern

Monthly sales show fluctuations over time.

Highest Monthly Sales

September 2018 = ₹589,384

Lowest Monthly Sales

February 2015 = ₹120,444

Interpretation

Sales vary considerably from month to month. The later years show several higher-sales months compared with the beginning of the dataset.

Forecasting Methods

Two forecasting methods can be applied:

1. Exponential Smoothing
2. ARIMA

22. Forecast Accuracy

Using the final six months as a test period:

Model	MAE	MSE	RMSE
Exponential Smoothing	102,290.57	17,284,857,373.73	131,471.89
ARIMA	103,309.53	17,461,213,846.83	132,140.89

Interpretation

Exponential Smoothing gives slightly lower MAE and RMSE in this evaluation.

Therefore, **Exponential Smoothing** can be preferred for this dataset when using these evaluation results.

23. Optimization Modeling

Business Problem

The supermarket wants to allocate promotional resources to products while maximizing expected profit.

Decision Variables

Let:

- A = promotional units allocated to high-profit category
 - B = promotional units allocated to second high-profit category
- Objective**

Maximize:

$$\text{Profit} = 40A + 30B$$

Constraints

$$2A + B \leq 100$$

$$A + 2B \leq 80$$

$$A, B \geq 0$$

Optimal Solution

$$A = 40 \text{ units}$$

$$B = 20 \text{ units}$$

$$\text{Maximum profit} = ₹2,200$$

Interpretation

The optimal solution uses the available promotional resources efficiently.

Note: This is a business optimization scenario constructed using the dataset's profitability context; the dataset itself does not contain resource-capacity variables.

24. Simulation Modeling

Business Problem

The supermarket needs to decide an appropriate daily inventory/order capacity.

The historical dataset has approximately **8 orders per observed day** on average.

A random daily demand simulation was created for 365 days.

Scenarios

Stock Capacity	Units/Orders Sold	Shortage	Service Level
5	1,778	1,146	60.8%

7	2,315	609	79.2%
9	2,659	265	90.9%

Result

The capacity of **9 orders per day** gives the highest simulated service level.

Interpretation

Increasing capacity reduces shortages and allows the supermarket to serve a larger proportion of daily demand.

25. ANOVA

Business Question

Does average Sales differ significantly among regions?

Groups

- North
- South
- East
- West
- Central

Hypotheses

H_0 : Mean Sales are equal across regions.

H_1 : At least one regional mean is different.

Results

- F-statistic = **0.295**
- p-value = **0.8288**
- α = **0.05**

Decision

Fail to reject H_0 .

Business Interpretation

There is no statistically significant evidence that average transaction Sales differs across regions.

Although West has the highest total sales, this is influenced by the number of transactions in each region.

26. Experimental Design

Business Objective

To study whether discount level and region affect Sales.

Factor 1

Region

Levels:

- North
- South
- East
- West
- Central

Factor 2

Discount Level

Levels:

- Low: $\leq 15\%$
- Medium: 16–25%
- High: $> 25\%$

Response Variable

Sales

Experimental Unit

Individual sales transaction.

Experiment

Transactions can be grouped according to Region and Discount Level to compare average Sales.

This design can help the business understand whether discount strategies work differently across regions.

27. Factorial Experiment

A two-factor factorial analysis was performed using:

Factor 1: Region

Factor 2: Discount Band

Response: Sales

Results

Effect	F-statistic	p-value	Decision
Region	0.302	0.8242	Not Significant
Discount Band	0.323	0.7243	Not Significant
Region × Discount	0.364	0.9020	Not Significant

Interpretation

Neither Region nor Discount Band has a statistically significant effect on Sales in this analysis.

The interaction between Region and Discount Band is also not significant.

Therefore, the dataset does not provide strong statistical evidence that the effect of discount differs across regions.

28. Business Insights and Findings

1. The dataset contains **9,994 supermarket transactions** across 11 variables.
 2. Total Sales are approximately **₹14.96 million**.
 3. Total Profit is approximately **₹3.75 million**.
 4. The overall profit margin is approximately **25.05%**.
 5. **Eggs, Meat & Fish** has the highest total Sales at approximately **₹2.27 million**.
 6. **Snacks** has the highest total Profit at approximately **₹568,179**.
 7. **West** is the highest-sales region with approximately **₹4.80 million** in Sales.
 8. **Health Drinks** is the highest-selling sub-category with approximately **₹1.05 million** in Sales.
 9. Sales and Profit have a positive correlation of **0.6053**.
 10. Discount has almost no linear correlation with Profit, with correlation approximately **0.0000**.
 11. Regression shows that Sales explains approximately **36.64%** of the variation in Profit.
 12. Approximately **50.08%** of transactions have Sales above the average Sales value.
 13. September 2018 recorded the highest monthly Sales of approximately **₹589,384**.
 14. The hypothesis test did not find a significant difference between average Sales and ₹1,500.
 15. ANOVA did not find a significant difference in average Sales among regions.
-

29. Business Recommendations

- Give greater attention to **Eggs, Meat & Fish** because it has the highest total Sales.
 - Promote **Snacks** because it generates the highest total Profit.
 - Maintain strong distribution and inventory availability in the **West region**.
 - Monitor the performance of lower-sales regions and identify opportunities for improvement.
 - Focus on high-performing sub-categories such as **Health Drinks and Soft Drinks**.
 - Avoid assuming that increasing discounts automatically increases Sales because the correlation analysis does not support this.
 - Use Sales-based forecasting for inventory and procurement planning.
 - Use Tableau dashboards to continuously monitor Sales, Profit, Region and Category performance.
 - Use statistical testing before making major changes to discount strategies.
 - Review monthly sales trends and prepare additional stock for high-sales periods.
-

30. Conclusion

This project analyzes the Supermart Grocery Sales dataset to understand sales performance, profitability, discount patterns, regional performance and sales trends over time.

The dataset contains 9,994 records and 11 columns. Data preprocessing and exploratory data analysis were used to examine the dataset. Descriptive statistics, correlation analysis, regression analysis and probability analysis were also performed.

The analysis indicates that Eggs, Meat & Fish generates the highest total Sales, Snacks generates the highest total Profit, and the West region has the highest total Sales. Sales and Profit have a positive correlation of approximately 0.6053.

The regression model indicates that Sales explains approximately 36.64% of the variation in Profit. The analysis also suggests that Discount has almost no linear correlation with Profit in this dataset.

The findings can help the supermarket understand product performance, monitor profitability and make better-informed decisions about promotions and inventory planning.

Overall, this project demonstrates how Business Analytics techniques can be applied to retail sales data to identify patterns, evaluate business performance and support data-driven decisions.

Overall, the project demonstrates how Business Analytics can help a supermarket make better decisions related to product performance, profitability, discount planning, forecasting and business strategy.

31. Appendix

Appendix A – Python Code

Include the complete Python code used in the Colab notebook for data loading, preprocessing, exploratory data analysis, statistical analysis, regression and other techniques actually performed.

Appendix B – Dataset

Provide the dataset name, original source URL and a brief description of the dataset. Include the CSV file separately if required by the college.

Appendix C – Data Preprocessing and EDA Outputs

Include screenshots of the actual Colab outputs for:

- First five records (`df.head()`)
- Dataset shape (`df.shape`)
- Column names (`df.columns`)
- Data types (`df.dtypes`)
- Missing-value check
- Duplicate-record check
- Descriptive statistics (`df.describe()`)

Appendix D – Charts and Visualizations

Include the charts actually generated in the notebook:

- Sales by Category
- Profit by Category
- Sales by Region
- Sales Distribution
- Profit Distribution
- Sales Box Plot
- Sales vs Profit Scatter Plot
- Correlation Heatmap

Appendix E – Tableau Dashboard

Insert the actual screenshot of the completed Tableau dashboard. Include the KPI cards, charts and filters used in the dashboard.

Appendix F – Statistical Analysis Outputs

Include the actual Colab outputs for the statistical techniques performed in the project, such as:

- Correlation matrix
- Regression results
- Probability calculations
- Sampling and sampling distribution

- Confidence interval estimation
- Bootstrapping
- Hypothesis testing
- A/B-style comparison
- Time-series forecasting and forecast accuracy
- Optimization modeling
- Simulation modeling
- ANOVA
- Experimental design and factorial analysis

Only include the techniques that were actually performed and whose results are available in the notebook.